Reliable Arrhythmia Detection from ECG

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Abstract

Automated electrocardiogram (ECG) classification systems are critical for the early detection of life-threatening cardiac arrhythmias. However, clinical datasets exhibit severe class imbalance, causing standard deep learning models to optimize for global accuracy at the expense of missing rare, dangerous conditions. In this paper, we develop a highly sensitive AI assistant to classify heartbeats into five diagnostic categories using the MIT-BIH Arrhythmia dataset, with a strict clinical focus on identifying Ventricular ectopic beats. To address the imbalance and penalize false negatives, we implement Cost-Sensitive Learning by injecting a custom severity multiplier into the loss function. We systematically compare a baseline Multilayer Perceptron (MLP) against a 1-Dimensional Convolutional Neural Network (1D-CNN) over five independent training runs. While both architectures achieved an exceptional mean Recall (0.96) for the Ventricular class, the MLP suffered from low mean Precision (0.48). Conversely, the 1D-CNN successfully improved that figure, nearly doubling the mean Precision to 0.86 and achieving a robust final mean Ventricular F1-Score of 0.91. Furthermore, gradient-based saliency maps confirm the CNN's superior interpretability, demonstrating highly focused attention on clinically relevant waveform segments. Our findings illustrate that coupling spatial feature extraction with domain-informed class weighting yields a robust screening tool that maximizes patient safety while minimizing alarm fatigue.

1 Introduction

Cardiovascular diseases remain a leading cause of mortality globally, making the early and accurate detection of cardiac arrhythmias a critical priority in medical diagnostics. The electrocardiogram (ECG) is the standard non-invasive tool used to monitor the heart's electrical activity. While human experts can effectively diagnose arrhythmias by analyzing the morphological patterns of an ECG waveform, manual continuous monitoring is highly labor-intensive and prone to human error due to fatigue. Consequently, there is a significant clinical need for automated, AI-driven ECG classification systems.

Despite the recent success of deep learning in time-series analysis, applying these models to medical datasets presents a distinct challenge: severe class imbalance. In clinical data, normal heartbeats naturally dominate the dataset, while life-threatening anomalies are rare. Standard neural networks, which optimize for global accuracy, often fall victim to the accuracy paradox. A model might achieve

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over 80% accuracy by simply classifying every beat as "Normal," effectively missing 100% of dangerous arrhythmias. In a medical context, the cost of a false negative (missing a fatal condition) far outweighs the cost of a false positive (triggering a false alarm). Among these anomalies, Ventricular ectopic beats are of paramount concern, as their frequent occurrence can quickly escalate into fatal conditions such as Ventricular Fibrillation.

1.1 Goals

To bridge the gap between mathematical optimization and clinical utility, this project proposes a cost-sensitive deep learning framework for heartbeat classification. We specifically prioritize maximizing the Sensitivity (Recall) and F1-Score of the rare, highly dangerous Ventricular class over global accuracy.

1.2 Approach

The main contributions of this work are as follows:

Domain-Informed Cost-Sensitive Learning: We implement a custom, severity-adjusted weight multiplier within the loss function, forcing the network to heavily penalize misclassifications of Ventricular beats.

Architectural Comparison: We systematically evaluate and compare the performance of a baseline Multi-Layer Perceptron (MLP) against a 1-Dimensional Convolutional Neural Network (1D-CNN), demonstrating the necessity of spatial feature extraction to reduce false-positive alarms.

Model Interpretability: We generate gradient-based saliency maps using backpropagation to visually validate that the neural networks are focusing on clinically relevant waveform morphologies (such as the QRS complex), rather than exploiting dataset noise.

2 Related Work

Traditional Machine Learning in ECG Analysis: Early approaches to automated ECG classification heavily relied on manual feature extraction and traditional machine learning algorithms. Methodologies utilizing standard time-series analysis for anomaly detection [1], rule-based fuzzy logic classifiers [2], and Support Vector Machines (SVMs) optimized via Genetic Algorithms [3] demonstrated foundational success. However, these techniques fundamentally depend on the manual engineering of morphological features (such as R-R intervals and QRS complex durations). Consequently, they are often brittle when exposed to standard clinical noise and struggle to scale effectively, generally yielding lower accuracy than modern representation learning techniques.

Deep Learning and the MIT-BIH Dataset: The paradigm shifted significantly with the advent of deep learning, which bypasses manual feature engineering by learning hierarchical representations directly from raw signals. Kachuee et al. [4] established a benchmark in this domain by preprocessing the MIT-BIH Arrhythmia database into a unified 1D-signal format and demonstrating that 1-Dimensional Convolutional Neural Networks (1D-CNNs) vastly outperform traditional models. Our work directly utilizes their standardized dataset split to build our baseline MLP and 1D-CNN architectures.

Addressing Class Imbalance: A persistent challenge in clinical deep learning is severe class imbalance. Recent works have attempted to mitigate this by manipulating the training data itself. For instance, recent applications of 1D-CNNs have leveraged resampling techniques, such as the Synthetic Minority Over-sampling Technique (SMOTE) and Gaussian Mixture Models, to artificially generate minority class samples [5]. In contrast to data-level synthesis, which risks introducing unnatural artifacts into medical time-series data, our approach employs an algorithm-level intervention. We implement domain-informed cost-sensitive learning to strictly prioritize Recall for dangerous arrhythmias without altering the underlying clinical data distribution.

Interpretability and Future Architectures: Frontier research of ECG classification integrates attention mechanisms to further refine spatial feature extraction. Recent frameworks utilizing 1D-CNNs augmented with attention mechanisms have achieved state-of-the-art F1-scores [6]. While such models cite the critical "black box" problem of deep learning in their motivation, achieving true visual interpretability remains a challenge. To bridge this gap, our work explicitly generates gradient-based saliency maps, providing the crucial visual evidence necessary for clinical trust. The integration of formal attention mechanisms alongside our cost-sensitive saliency mapping represents a highly promising avenue for future work.

3 Dataset

Source and Preprocessing: This project utilizes the MIT-BIH Arrhythmia Database, originally hosted on PhysioNet, which contains annotated ambulatory ECG recordings. Specifically, we use the pre-processed and segmented version of the dataset provided by Kachuee et al. [4]. In this iteration, continuous ECG signals were segmented into individual heartbeats, padded or cropped to a fixed length, and their amplitudes were normalized to a range of $[0, 1]$.

Data Dimensions: The complete dataset consists of 109,446 individual heartbeat samples. Each input sample is a 1-Dimensional time-series array of length 187, representing a single heartbeat sampled at a frequency of 125 Hz. The target output is a single discrete integer representing one of five diagnostic categories.

Diagnostic Categories: The heartbeats are classified into five distinct groups based on the Association for the Advancement of Medical Instrumentation (AAMI) standard:

- **Class 0 (N):** Non-ectopic (Normal) beats.
- **Class 1 (S):** Supraventricular ectopic beats.
- **Class 2 (V):** Ventricular ectopic beats (Our primary clinical focus).
- **Class 3 (F):** Fusion beats.
- **Class 4 (Q):** Unknown or unclassifiable beats.

Dataset Splits and Class Imbalance: The dataset is pre-divided into a training set of 87,554 samples and a strictly hold-out testing set of 21,892 samples. A defining characteristic of this clinical dataset is its severe class imbalance. The 'Normal' (N) class dominates the distribution, accounting for approximately 82.8% of the training data. Conversely, the dangerous 'Ventricular' (V) class constitutes only 6.6% of the data, and 'Fusion' (F) beats account for less than 1%. This extreme skew necessitates the specialized cost-sensitive evaluation metrics and loss weighting strategies detailed in the subsequent sections.

Table 1: Distribution of Heartbeat Classes in the Training and Testing Sets

Class ID	Diagnostic Category	Training Samples	Testing Samples
0	N (Normal)	72,471	18,118
1	S (Supraventricular)	2,223	556
2	V (Ventricular)	5,788	1,448
3	F (Fusion)	641	162
4	Q (Unknown)	6,431	1,608
Total		**87,554**	**21,892**

4 Methodology

To establish the importance of spatial and morphological feature extraction in electrocardiogram analysis, we formulate our approach as a comparative study between a standard Multi-Layer Perceptron (MLP) and a 1-Dimensional Convolutional Neural Network (1D-CNN). Both architectures are trained to output raw logits across the five diagnostic categories, optimized via a cost-sensitive `CrossEntropyLoss` function.

4.1 Baseline Architecture: Multi-Layer Perceptron (MLP)

The MLP serves as our baseline model, processing the heartbeat as a flattened, discrete vector of amplitudes. The input layer accepts the raw 187 time-step ECG signal. The network comprises three fully connected hidden layers with a progressively decreasing dimensionality of 128, 64, and 32 neurons. This funnel-like structure is designed to compress the feature space and extract salient patterns. To introduce non-linearity, Rectified Linear Unit (ReLU) activation functions follow each linear transformation. To mitigate the risk of overfitting on the training data, Dropout regularization ($p = 0.2$) is applied after each hidden layer. The final output layer maps the 32-dimensional feature vector directly to the 5 diagnostic classes.

4.2 Proposed Architecture: 1-Dimensional Convolutional Neural Network (1D-CNN)

While the MLP assesses global amplitude correlations, it inherently discards the temporal topology of the ECG waveform. To explicitly capture the local morphological structures, such as the width of the QRS complex or the inversion of the T-wave, we implement a 1D-CNN.
The input is reshaped to a single-channel sequence of shape $(B, 1, 187)$, where B represents the batch size, allowing the convolutional filters to slide across the temporal axis. The feature extraction backbone consists of three sequential convolutional blocks. Each block applies a `Conv1d` layer, followed by Batch Normalization to stabilize internal covariate shift and accelerate training, a ReLU activation, and a `MaxPool1d` operation (kernel size of 2). The number of convolutional filters increases progressively ($32 \rightarrow 64 \rightarrow 128$) to capture increasingly abstract hierarchical features, while the pooling layers systematically compress the temporal sequence length from 187 down to approximately 23 time-steps.
Following the convolutional blocks, the resulting spatio-temporal feature maps are flattened into a one-dimensional tensor and passed into a fully connected classifier network. This dense network consists of two hidden layers (256 and 128 neurons). Given the substantial increase in trainable parameters at the flattening boundary, we apply a more aggressive Dropout regularization ($p = 0.3$) between the dense layers to ensure robust generalization. The network terminates in a 5-neuron linear layer yielding the class logits.

Table 2: Architectural details of the baseline MLP and the proposed 1D-CNN models.

Table 2a: MLP Architecture

Layer	Components	Output Shape
Input	Raw Signal	187
Hidden 1	Linear, ReLU, Dropout (0.2)	128
Hidden 2	Linear, ReLU, Dropout (0.2)	64
Hidden 3	Linear, ReLU, Dropout (0.2)	32
Output	Linear (Logits)	5

Table 2b: 1D-CNN Architecture

Block / Layer	Components	Channels / Features
Input	Reshape to $(1, 187)$	1
Conv Block 1	Conv1d, BatchNorm, ReLU, Pool	32
Conv Block 2	Conv1d, BatchNorm, ReLU, Pool	64
Conv Block 3	Conv1d, BatchNorm, ReLU, Pool	128
Flatten	Flatten Tensor	2944
Dense 1	Linear, ReLU, Dropout (0.3)	256
Dense 2	Linear, ReLU, Dropout (0.3)	128
Output	Linear (Logits)	5

4.3 Architectural Rationale and Regularization Strategy

The structural dimensions of both architectures were intentionally designed to gradually compress information, forcing the networks to learn generalizable patterns rather than memorizing noise.

For the baseline MLP, we implemented a standard "information funnel" design. By progressively halving the dimensionality of the hidden layers ($128 \rightarrow 64 \rightarrow 32$), we force the network to discard irrelevant amplitude fluctuations and encode only the most critical global features of the heartbeat before the final classification layer. A standard dropout rate of 0.2 was sufficient to provide baseline regularization for this relatively compact parameter space.

In contrast, the 1D-CNN was designed for hierarchical spatial feature extraction. As the signal passes through the convolutional blocks, the temporal length is systematically compressed by the pooling layers, while the channel depth is expanded ($32 \rightarrow 64 \rightarrow 128$). This allows the early layers to detect low-level, rapid variations (such as the sharp R-peak), while deeper layers combine these into high-level morphological concepts (such as the entire QRS complex).

Justification for Asymmetric Dropout: A critical divergence in our regularization strategy is the application of a more aggressive dropout rate (0.3) in the CNN's fully connected classifier, compared to the MLP (0.2). This is necessitated by the architectural bottleneck that occurs immediately after the convolutional backbone. Flattening 128 channels across the remaining spatial dimensions results in a dense vector of 2,944 features. Mapping this vector to the first 256-neuron dense layer creates a massive parameter explosion, resulting in over 750,000 trainable weights in a single layer. Networks with such dense parameter concentrations are highly susceptible to overfitting training data. The elevated 0.3 dropout rate acts as a strict countermeasure, shutting off 30% of the neurons during each training step. This forces the CNN to learn robust, redundant representations of the ECG morphology rather than relying on highly specific, brittle feature pathways.

4.4 Training Protocol and Hyperparameters

Both models were trained using the PyTorch framework. We optimized the network parameters using the Adam (Adaptive Moment Estimation) optimizer, initialized with a standard learning rate of 0.001 to ensure stable convergence. The dataset was processed in mini-batches of 64 samples. Through trial and error, we established 10 to be the right number of training epochs, which consistently led to convergence towards a stable minima while also preventing overfitting.

4.5 Cost-Sensitive Loss Formulation

The most critical methodological design of this project addresses the severe clinical class imbalance. Standard classification tasks assume equal cost for all misclassifications. However, in our clinical context, falsely predicting a 'Normal' beat when the true condition is 'Ventricular' constitutes a life-threatening false negative.

To counteract the model's natural bias toward the majority class, we employed Cost-Sensitive Learning by heavily modifying the standard PyTorch `CrossEntropyLoss` criterion. First, we calculated the mathematically balanced inverse-frequency weight for each class c:

$$w_c = \frac{N_{total}}{C \times N_c}$$

where N_{total} is the total number of training samples, C is the number of diagnostic categories (5), and N_c is the number of samples in class c.

While this normalizes the dataset mathematically, it is blind to clinical severity. For instance, assigning the highest penalty to the rarest but less immediately lethal 'Fusion' class. Therefore, we injected specific domain knowledge by applying a manual clinical severity multiplier (λ) strictly to the Ventricular class (V):

$$w_V^{clinical} = w_V \times \lambda$$

Through iterative experimentation, we empirically set the clinical multiplier to $\lambda = 5.0$. This specific value was selected because it yielded a good balance during early trials, elevating the penalty for misclassifying a Ventricular beat just enough to ensure near-perfect Sensitivity without causing a complete collapse in Precision. This customized weight tensor was passed directly into the loss function, forcing the gradient descent algorithm to prioritize Ventricular detection over global accuracy. However, we explicitly acknowledge a critical limitation in this hyperparameter selection: the exact magnitude of λ fundamentally represents a medical threshold rather than a purely mathematical one. In a true clinical deployment, this value cannot be optimized solely via empirical

grid search. It must be strictly calibrated through interdisciplinary consultation with cardiologists to accurately quantify the relative mortality risk of a missed Ventricular beat against the operational dangers of false-positive alarm fatigue.

5 Evaluation

5.1 Evaluation Methodology and Statistical Robustness

Evaluating deep learning architectures on a single training instance is inherently susceptible to variance caused by random weight initialization and stochastic mini-batch shuffling. Relying on a single fixed random seed ensures code reproducibility but risks reporting an unrepresentative outlier: either artificially inflating or deflating the perceived performance of the model.

To ensure statistical rigor and to validate the stability of our cost-sensitive learning approach, we adopt a multi-run evaluation strategy. Both the baseline MLP and the 1D-CNN were independently trained and evaluated across five distinct random initializations. The primary metrics of interest (outlined below) are reported as the mean value across these five runs, accompanied by their standard deviation. This approach provides a robust estimate of the models' true generalization capabilities. For qualitative visual analysis, including Confusion Matrices and gradient-based Saliency Maps, we present visualizations derived from a single representative run whose metrics most closely align with the calculated statistical mean.

5.2 Evaluation Metrics

Due to the extreme class imbalance present in the MIT-BIH dataset, standard global accuracy is insufficient and highly misleading. A model predicting the majority 'Normal' class exclusively would achieve a baseline accuracy of over 80% while entirely failing its clinical objective. Therefore, our evaluation is anchored on the following metrics, computed specifically for the Ventricular (V) class:

- **Sensitivity/Recall:** The proportion of actual Ventricular beats successfully identified by the model. In a clinical screening context, maximizing this metric is paramount to prevent fatal false negatives.
- **Precision:** The proportion of predicted Ventricular beats that were truly Ventricular.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a single balanced metric to evaluate the trade-off between missing dangerous conditions and triggering false alarms.

5.3 Quantitative Results

Table 3 presents the statistical performance of the MLP and 1D-CNN architectures over five independent training runs. The cost-sensitive weighting mechanism successfully achieved the primary clinical objective across both models, with the MLP and 1D-CNN yielding an exceptional mean Ventricular Recall of 0.96. This confirms that the severity multiplier of 5.0 consistently forces the network to minimize life-threatening false negatives, regardless of weight initialization.However, the architectural limitations of the MLP are evident in its low mean Precision (0.48 ± 0.06). Without spatial awareness, the MLP triggers a high volume of false positives. The 1D-CNN decisively resolves this trade-off. By leveraging convolutional filters to extract local morphological features, the CNN nearly doubles the mean Precision to 0.86 ± 0.04 while maintaining the high Recall baseline. Consequently, the 1D-CNN achieves a superior mean Ventricular F1-Score of 0.91 ± 0.02, proving it to be a significantly more robust and reliable clinical screening tool.

Table 3: Performance metrics for the Ventricular class across 5 independent runs (Mean $\pm$ Std).

Architecture	Precision	Recall (Sensitivity)	F1-Score	Overall Accuracy
Baseline MLP	0.48 ± 0.06	0.96 ± 0.01	0.64 ± 0.05	0.81 ± 0.02
Proposed 1D-CNN	$\mathbf{0.86 \pm 0.04}$	$\mathbf{0.96 \pm 0.01}$	$\mathbf{0.91 \pm 0.02}$	$\mathbf{0.94 \pm 0.02}$

5.4 Qualitative Diagnostics and Interpretability

To visually validate the quantitative metrics, we extracted diagnostic plots from a representative training instance, whose performance closely mirrors the statistical mean (CNN F1-Score: 0.91).

The confusion matrices (Figure 2) illustrate the stark reduction in false positives achieved by the CNN compared to the MLP. Furthermore, tracking the Ventricular F1-score across the training epochs (Figure 1) demonstrates the CNN's rapid convergence and stability, whereas the MLP exhibits higher variance.

Finally, to address the "black box" nature of deep learning, we generated gradient-based saliency maps using backpropagation (Figure 1). A large gradient indicates that a specific segment of the ECG signal had a significant influence on the model's Ventricular classification. The CNN demonstrates highly refined, localized attention directly centered on the abnormal QRS complex. In contrast, the MLP's attention is noisy and widely distributed across the flat resting phase of the heartbeat, visually confirming its inability to isolate critical temporal structures.

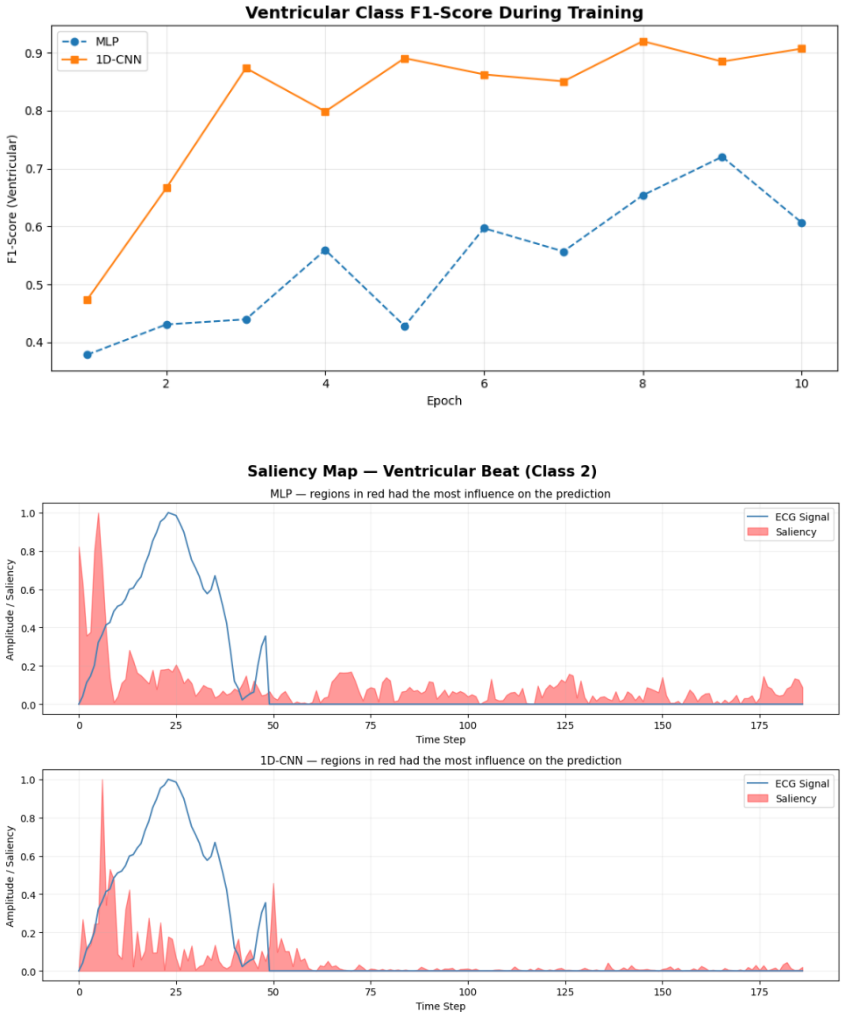


Figure 1: Top: Validation F1-Score progression across training epochs. Bottom: Gradient-based saliency maps demonstrating the CNN's superior focus on the QRS complex compared to the MLP.

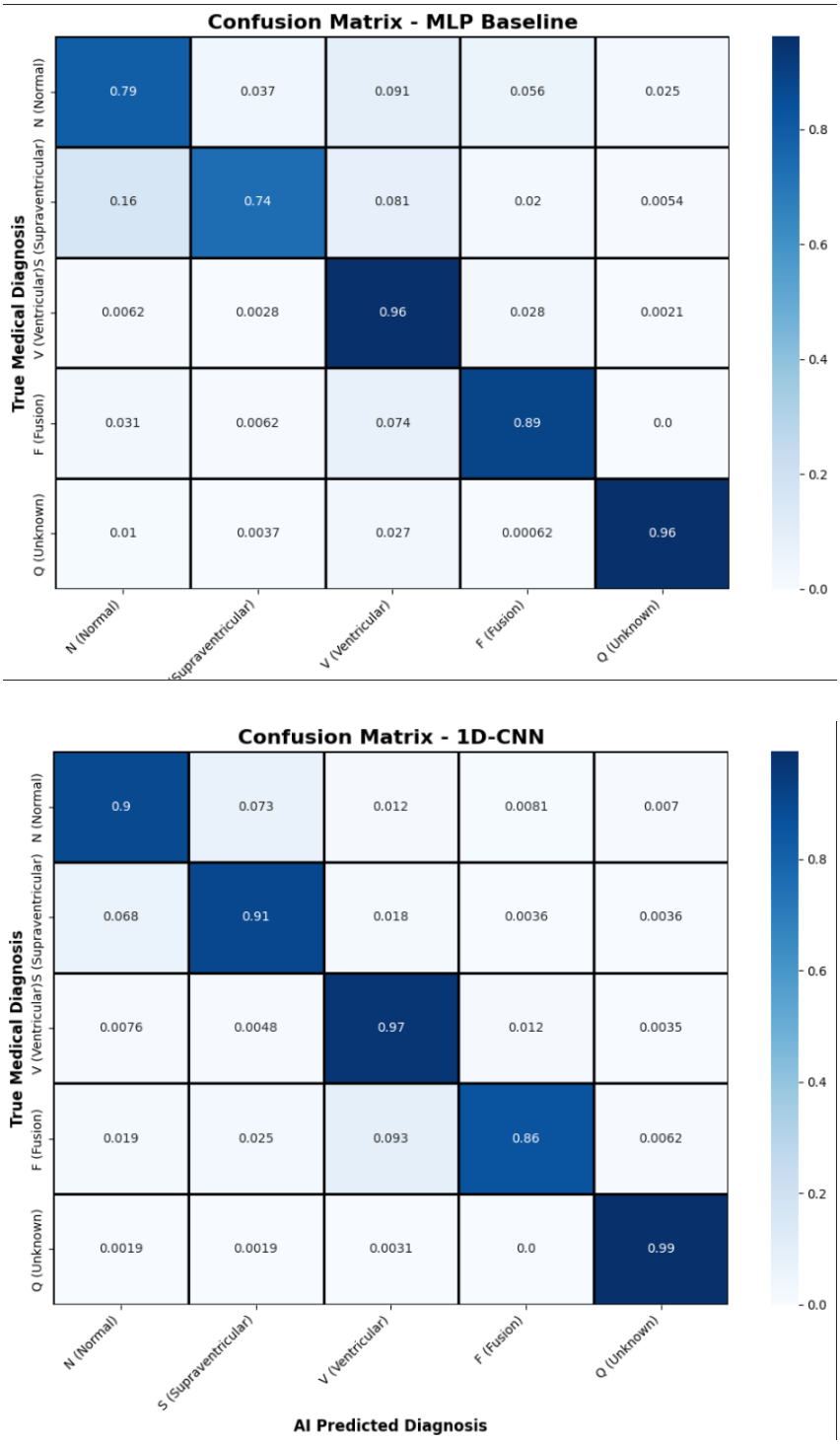


Figure 2: Confusion matrices from the representative run for the MLP (top) and 1D-CNN (bottom). The CNN drastically reduces false positives while maintaining high sensitivity for the Ventricular class.

6 Discussion

6.1 Interpretation of Results and Interpretability

The primary objective of this study was to develop an automated ECG classification framework that rigidly prioritizes patient safety by minimizing false-negative predictions of life-threatening Ventricular arrhythmias. Our results demonstrate that algorithm-level intervention, specifically, domain-informed Cost-Sensitive Learning, is highly effective at achieving this goal. By injecting a severity multiplier into the loss function, both the baseline MLP and the proposed 1D-CNN achieved an exceptional mean Recall of 0.96.

However, the comparative analysis sharply highlights the necessity of spatial feature extraction in time-series medical data. The MLP, relying strictly on global amplitude correlations, suffered from a high false-positive rate (Precision: 0.48). Conversely, the 1D-CNN successfully captured the local temporal topology of the heartbeat, doubling the Precision to 0.86 and yielding a highly robust F1-Score of 0.91. This quantitative superiority is explicitly validated by our representative saliency maps. The visual evidence confirms that the CNN achieves its high precision by consistently anchoring its attention to the abnormal QRS complex, whereas the MLP's attention remains noisy and highly distributed across clinically irrelevant resting phases.

6.2 Clinical Trade-offs and Limitations

While our model achieves high diagnostic sensitivity, it does so at the cost of overall accuracy and precision. We empirically set the clinical multiplier ($\lambda = 5.0$) to heavily bias the network toward over-sensitivity. In a real-world Intensive Care Unit, a high volume of false positives inevitably leads to "alarm fatigue," a dangerous phenomenon where medical staff become desensitized to continuous alerts, posing its own unique risk to patient safety. Therefore, a critical limitation of this work is the manual, empirical tuning of λ. Moving this system to deployment requires interdisciplinary consultation with cardiologists to mathematically calibrate the exact operational threshold between catching every fatal arrhythmia and maintaining a manageable alarm rate.

6.3 Future Work

Looking forward, the absolute cutting edge of sequence modeling utilizes attention-based architectures. While our 1D-CNN effectively localized the QRS complex, integrating formal Attention Mechanisms [6] could allow the network to dynamically weight the importance of different temporal segments in real-time. This would not only push the F1-Score closer to the theoretical maximum but would also inherently generate built-in attention maps, providing doctors with an even more rigorous, native form of visual interpretability.

7 Appendix



Figure 3: Additional saliency samples: It is clearly visible, how the CNN ignores the empty signal after the heart beat.

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